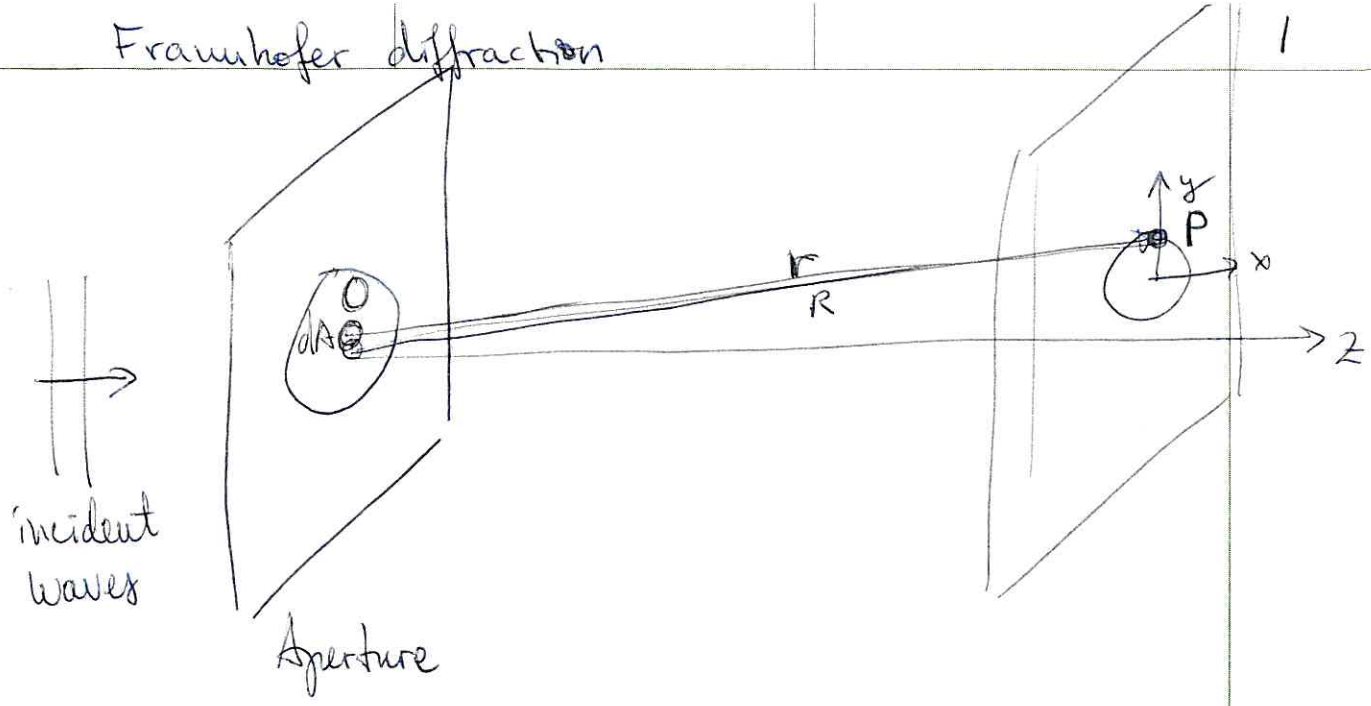
Fraunhofer diffraction



$\vec{R}$ = coordinate of P

$r = |\vec{R} - \vec{r}\,'|$ = distance from dA to P

$\vec{r}\,'$ = coordinate of element dA

$$dE(P) = \underbrace{\frac{E_0 e^{-i\omega t} dA'}{\lambda}}_{\substack{\text{source strength} \\ \text{(field at aperture)}}} \cdot \underbrace{\frac{e^{ikr}}{r}}_{\substack{\text{Huygens} \\ \text{spherical wave}}}$$

(solution of 3d wave equation)

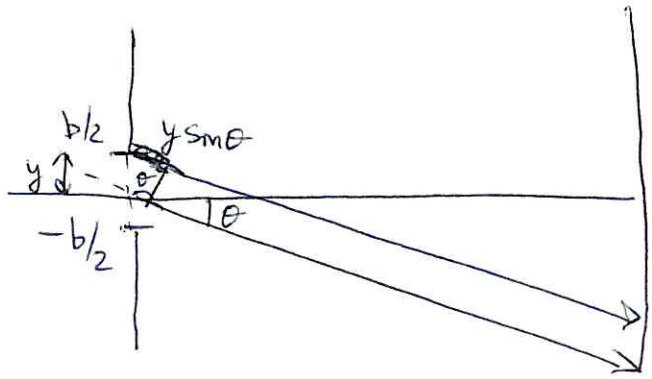
$$\Rightarrow E(P) = (\text{const}) \int_{(\text{aperture})} \frac{e^{ikr}}{r}\, dA$$

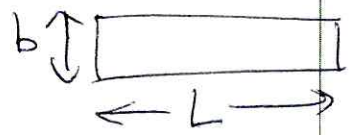
Far-field diffraction when plane wave approximation works well $\Rightarrow \vec{k}\cdot\vec{r} = kr$

for $r \gg \lambda$ (so that $kr \gg 1$) can neglect variation of r in factor $\frac{1}{r}$

$$\Rightarrow E(P) \sim \int e^{ikr}\, dA$$

The single slit





$$dA = L\,dy$$

$$r = r_0 + y\sin\theta$$

$$E(P) = C\, e^{ikr_0} L \int_{-b/2}^{b/2} e^{iky\sin\theta}\, dy =$$

$$= C\, e^{ikr_0} L \left. \frac{e^{iky\sin\theta}}{k\sin\theta} \right|_{-b/2}^{b/2} =$$

$$= C e^{ikr_0} L\, b\, 2i \frac{\sin\left(\frac{b}{2} k\sin\theta\right)}{\frac{b}{2} k\sin\theta} =$$

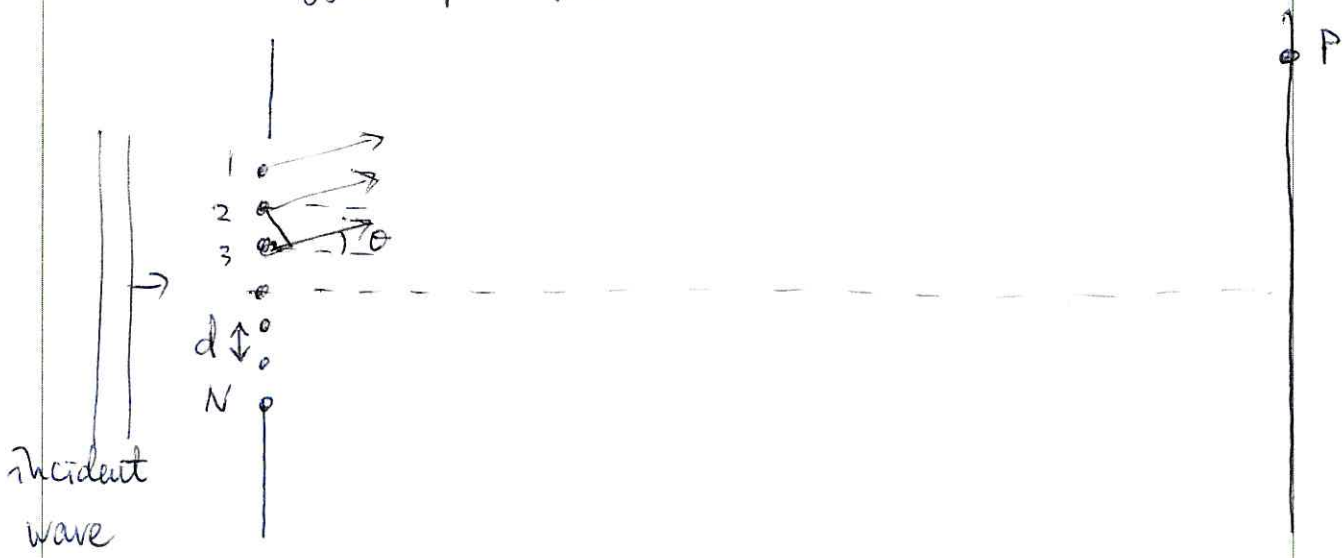
$$= C' \frac{\sin\left(\frac{\pi b\sin\theta}{\lambda}\right)}{\frac{\pi b\sin\theta}{\lambda}}; \qquad C' = i C e^{ikr_0} L b$$

$$\frac{b}{2} k\sin\theta = \frac{b}{2}\frac{2\pi}{\lambda}\sin\theta = \frac{\pi b\sin\theta}{\lambda}$$

$$\Rightarrow \text{intensity} \sim |E(P)|^2 = I_0 \left(\frac{\sin\beta}{\beta}\right)^2$$

where $\beta = \frac{1}{2} k b \sin\theta = \pi b \sin\theta / \lambda$.

Via Huygens principle:



Approximate single slit of width D by N narrow slits of width $\frac{D}{N-1}$
They radiate coherently.

$$d = D/_{N-1}$$

$$E_p = A(r)\cos(kr_1 - \omega t) + A(r)\cos(kr_2 - \omega t) + \ldots + A(r)\cos(kr_N - \omega t$$

$$r_2 = r_1 + d\sin\theta$$
$$r_3 = r_2 + d\sin\theta = r_1 + 2d\sin\theta$$
$$\vdots$$
$$r_N = r_{N-1} + d\sin\theta = r_1 + d\sin\theta\,(N-1)$$

$$\cos(kr - \omega t) = \mathrm{Re}\, e^{i(kr - \omega t)}$$

$$\Rightarrow E_p = A(r)\,\mathrm{Re}\left\{ e^{-i\omega t}\left[e^{ikr_1} + e^{ikr_2} + \ldots + e^{ikr_N}\right]\right\}$$

$$\mathrm{Re}\left\{ e^{-i\omega t} e^{ikr_1}\left[1 + e^{i\Delta\varphi} + e^{i2\Delta\varphi} + \ldots + e^{i(N-1)\Delta\varphi}\right]\right\}$$

where $\Delta\varphi = k(r_2 - r_1) = kd\sin\theta = \frac{2\pi d}{\lambda}\sin\theta$

Geometric series

$$S = \sum_{n=0}^{N-1} x^n = 1 + x + x^2 + \dots + x^{N-1}$$

$$xS = \sum_{n=0}^{N-1} x^{n+1} = x + x^2 + x^3 + \dots + x^{N-1} + x^N = x^N + (S-1)$$

$$\Rightarrow (x-1)S = x^N - 1$$

$$S = \frac{x^N - 1}{x-1} = \frac{e^{iN\Delta\varphi} - 1}{e^{i\Delta\varphi} - 1} = \frac{e^{i\frac{N}{2}\Delta\varphi}\left(e^{i\frac{N}{2}\Delta\varphi} - e^{-i\frac{N}{2}\Delta\varphi}\right)}{e^{i\frac{\Delta\varphi}{2}}\left(e^{i\frac{\Delta\varphi}{2}} - e^{-i\frac{\Delta\varphi}{2}}\right)}$$

$$= e^{i\frac{1}{2}(N-1)\Delta\varphi} \frac{2i\sin\left(\frac{N}{2}\Delta\varphi\right)}{2i\sin\left(\frac{\Delta\varphi}{2}\right)}$$

$$\Rightarrow E_p = A(r)\, e^{-i\omega t}\, e^{ikr_1 + \frac{i}{2}(N-1)kd\sin\theta} \frac{\sin\frac{1}{2}N\Delta\varphi}{\sin\frac{1}{2}\Delta\varphi}$$

but $r_1 + \frac{1}{2}(N-1)d\sin\theta = r$ = distance from the mid-point of the slit to the point P!

$$\Rightarrow E_p = A(r) \frac{\sin\left(\frac{N\pi d\sin\theta}{\lambda}\right)}{\sin\left(\frac{\pi d\sin\theta}{\lambda}\right)} \cdot \cos(kr - \omega t).$$

<u>Check</u> let $N=2$ then $\sin\left(2\,\frac{\Delta\varphi}{2}\right) = 2\sin\frac{\Delta\varphi}{2}\cos\frac{\Delta\varphi}{2}$

$$\Rightarrow E_p(N=2) = \underbrace{2A(r)\cos\frac{\Delta\varphi}{2}}_{2A\cos\left(\frac{\pi d\sin\theta}{\lambda}\right)} \cdot \cos(kr - \omega t)$$

$$\frac{\pi d\sin\theta}{\lambda} = \begin{cases} \pi n, & \text{constructive} \\ (2n+1)\frac{\pi}{2}, & \text{destructive} \end{cases}$$

$$\Rightarrow \left.\begin{array}{ll} d\sin\theta = n\lambda & \text{(constructive)} \\ d\sin\theta = (n'+\frac{1}{2})\lambda & \text{(destructive)} \end{array}\right\} \quad \underline{\text{OK}}$$

Diffraction from single slit of width D

$N \to \infty$, $d \to 0$ limit so that $\underline{Nd = D}$

$$\Rightarrow Nd\,\frac{\pi\sin\theta}{\lambda} = D\,\frac{\pi\sin\theta}{\lambda}$$

$$\pi d\,\frac{\sin\theta}{\lambda} = D\,\frac{\pi\sin\theta}{\lambda}\cdot\frac{1}{N}$$

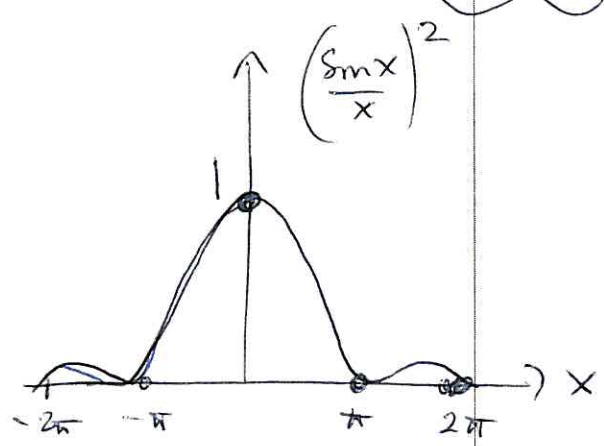
$$\Rightarrow \frac{\sin\left(N\pi d\sin\theta/\lambda\right)}{\sin\left(\pi d\sin\theta/\lambda\right)} = \frac{\sin\left(\pi D\sin\theta/\lambda\right)}{\sin\left(\frac{\pi D\sin\theta}{\lambda N}\right)} \Rightarrow$$

$$\sin\left(\frac{\phi}{N}\right) \approx \frac{\phi}{N} \quad (N\to\infty)$$

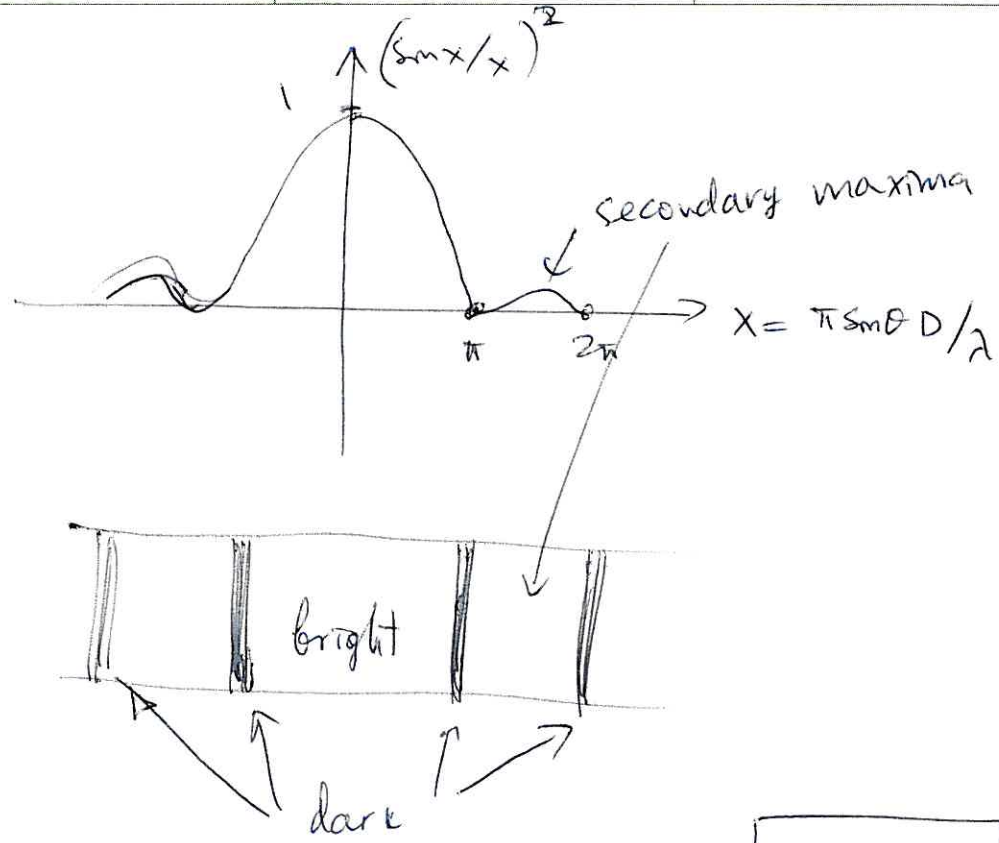
$$\Rightarrow E_p = \underbrace{\underbrace{N A(r)}_{\text{finite } N\to\infty \text{ limit}}}_{\tilde{A}(r)} \frac{\sin\left(\pi D\sin\theta/\lambda\right)}{\left(\pi D\sin\theta/\lambda\right)}$$

agrees with more formal calculation on pages 1, 2

$$\Rightarrow I = I_0\,\frac{\sin^2\left(\frac{\pi D\sin\theta}{\lambda}\right)}{\left(\frac{\pi D\sin\theta}{\lambda}\right)^2}$$

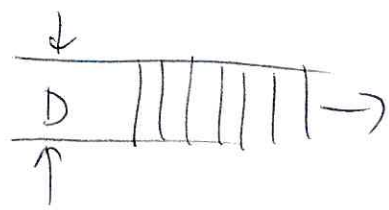


$$I(x) = 0 \text{ for } x = \frac{\pi D\sin\theta}{\lambda} = m\pi \quad (m \in \mathbb{Z})$$



First minimum: $x = \pi \Rightarrow \boxed{\sin\theta = \frac{\lambda}{D}}$

<u>Also!</u> Angular spread of the beam of width D:



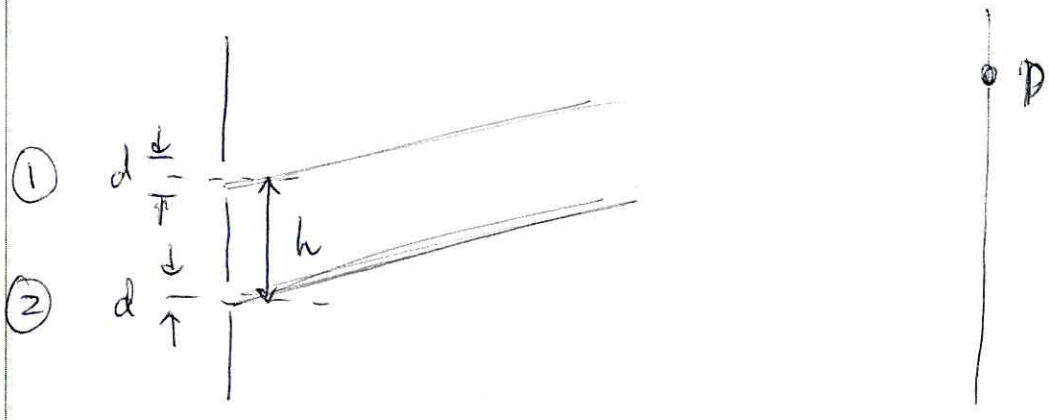
by Huygens principle any vertical crossection can be considered as a secondary source

$$\Rightarrow I(\theta) = I_0 \left[\frac{\sin\left(\frac{1}{2} k D \sin\theta\right)}{\frac{1}{2} k D \sin\theta}\right]^2$$

is the answer for this problem as well

$$\Rightarrow \Delta\theta = \frac{\lambda}{D}$$

$N=2$ <u>Two slits</u> (of width d each) separated by h



el. field from each slit is given by

$$E_{slit} = \tilde{A}(r) \frac{\sin(\pi d \sin\theta/\lambda)}{\pi d \sin\theta/\lambda} \cos(kr - \omega t)$$

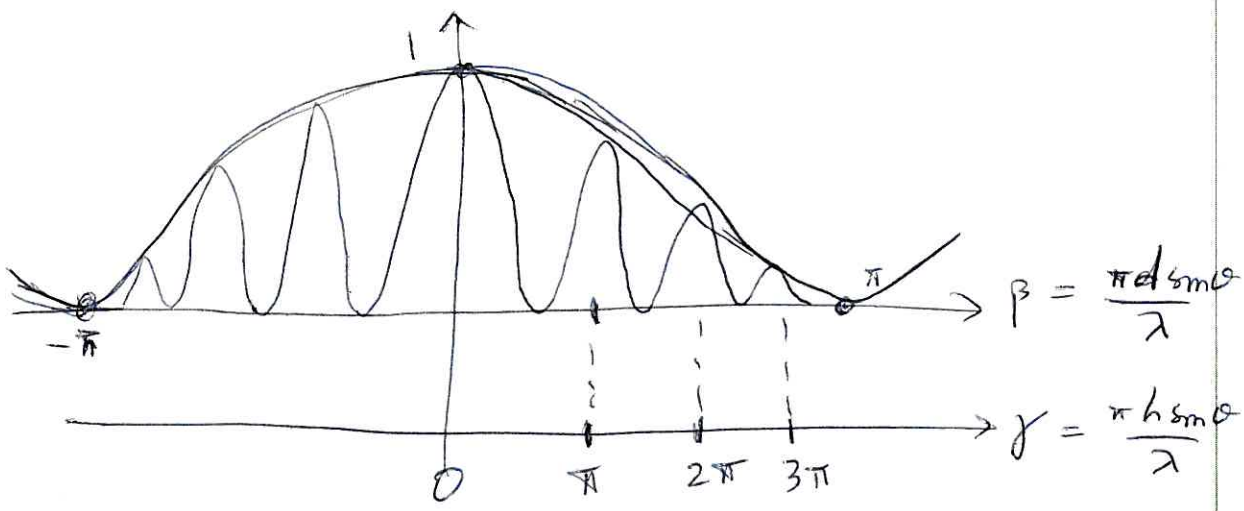
but the centers of the two slits are now shifted by h, $\Rightarrow$ there is additional phase difference $\cos\left(\frac{1}{2}\Delta\gamma\right)$ where $\Delta\gamma = kh\sin\theta = \frac{2\pi}{\lambda} h \sin\theta$

$$\Rightarrow \text{result } E_p = E_1 + E_2 = \tilde{A}(r) \frac{\sin\left(\frac{\pi d \sin\theta}{\lambda}\right)}{\frac{\pi d \sin\theta}{\lambda}} \cdot \cos\left(\frac{\pi h \sin\theta}{\lambda}\right)$$

and $I = |E_p|^2$.

For <u>$h > d$</u> the result is Young's interference pattern modulated by $\left(\frac{\sin\beta}{\beta}\right)^2$ factor due to finite width of a given slit.

$$\left(\beta = \frac{\pi d \sin\theta}{\lambda}\right)$$



$$I = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \cos^2(\gamma)$$

min for $\beta = \pi n$

$\sin\theta = \frac{\lambda}{d} n$

max for $\gamma = \pi n'$

$\sin\theta = \frac{\lambda}{h} n'$

For $h \gg d$ there will be many maxima of $\cos^2(\gamma)$ within the first broad peak of $\left(\frac{\sin\beta}{\beta}\right)^2$!

(their number $= n' \sim \frac{h}{d}$).

Grating

N slits (each of width d) separated by h from each other

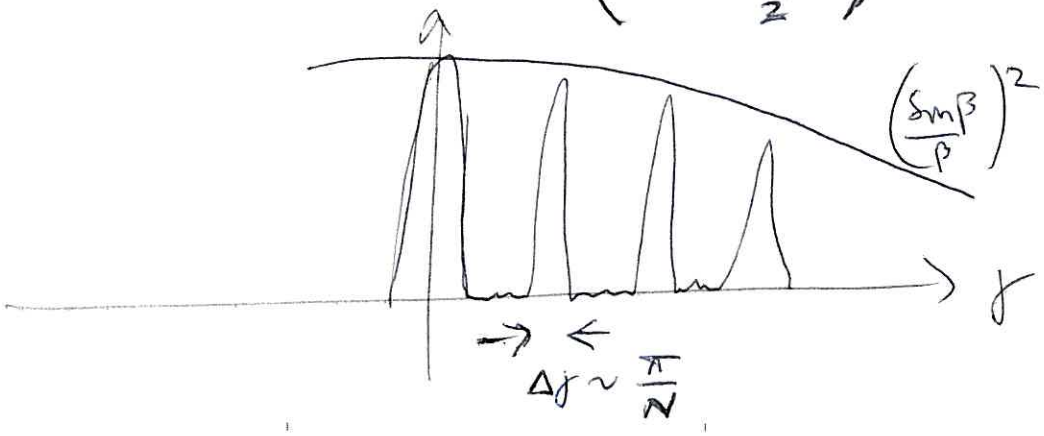
$$E_p = E_1 + E_2 + \ldots + E_N = \tilde{A}(r)\left(\frac{\sin\beta}{\beta}\right)\left[\frac{\sin\left(\frac{N}{2}\Delta\gamma\right)}{\sin\frac{\Delta\gamma}{2}}\right]\cdot \cos(\omega t - kr)$$

$$\beta = \frac{\pi d \sin\theta}{\lambda}$$

$$\frac{1}{2}\Delta\gamma = \frac{\pi h \sin\theta}{\lambda}$$

$\Rightarrow$ Intensity of such diffraction grating is

$$I = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \left(\frac{\sin\left(N\frac{\Delta\gamma}{2}\right)}{N\sin\frac{\Delta\gamma}{2}}\right)^2$$



$$\left(\frac{\sin N\gamma}{N\sin\gamma}\right)^2 = f(\gamma)$$

$$f(\gamma\to 0)\to 1$$

$$f(\gamma\to\pi) = \left(\frac{e^{iN(\pi-\varepsilon)} - e^{-iN(\pi-\varepsilon)}}{N\left[e^{i(\pi-\varepsilon)} - e^{-i(\pi-\varepsilon)}\right]}\right)^2 =$$

$\gamma = \pi - \varepsilon$

$$= \left(\frac{e^{-iN\varepsilon} - e^{iN\varepsilon}}{N\left[e^{-i\varepsilon} - e^{i\varepsilon}\right]}\right)^2 = \left(\frac{\sin N\varepsilon}{N\sin\varepsilon}\right)^2 = 1$$

$\varepsilon\to 0$

$\Rightarrow$ maxima occur for $\frac{\gamma}{2} = \pi m$

$$\frac{\pi h \sin\theta}{\lambda} = \pi m$$

$$\boxed{\sin\theta = \frac{\lambda}{h} m}$$

And these maxima are very sharp

$$\delta\left(\frac{N\gamma}{2}\right) = \pi \Rightarrow N\frac{\pi h}{\lambda}\delta(\sin\theta) = \pi$$

$$N\frac{\pi h}{\lambda}\cos\theta\,\delta\theta = \pi$$

$$\delta\theta = \frac{\lambda}{Nh\cos\theta} \xrightarrow[N\to\infty]{} 0$$